

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Its temperature increases [to point Y] (1) and then [from Y to white dwarf] its temperature decreases (1)		2		2		
		(ii)		Its brightness decreases or it gets dimmer		1		1		
	(b)			[Main sequence →] super giant (1) → supernova (1) → black hole or neutron star (1)	3			3		
	(c)			$8.6 \times 63\,241 = 543\,872.6$ [AU] (1) $543\,872.6$ (ecf) $\times 1.5 \times 10^8 = 8.158 \times 10^{13}$ [km] (1) $8.158 \times 10^{13} \times 1\,000$ (1 - conversion) = 8.158×10^{16} [m] For an answer of 8.2×10^n where n is not equal to 16 award 2 marks		3		3	3	
				Question 5 total	3	6	0	9	3	0